



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 1: Course Introduction

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About myself

Yajing SHEN (申亞京)

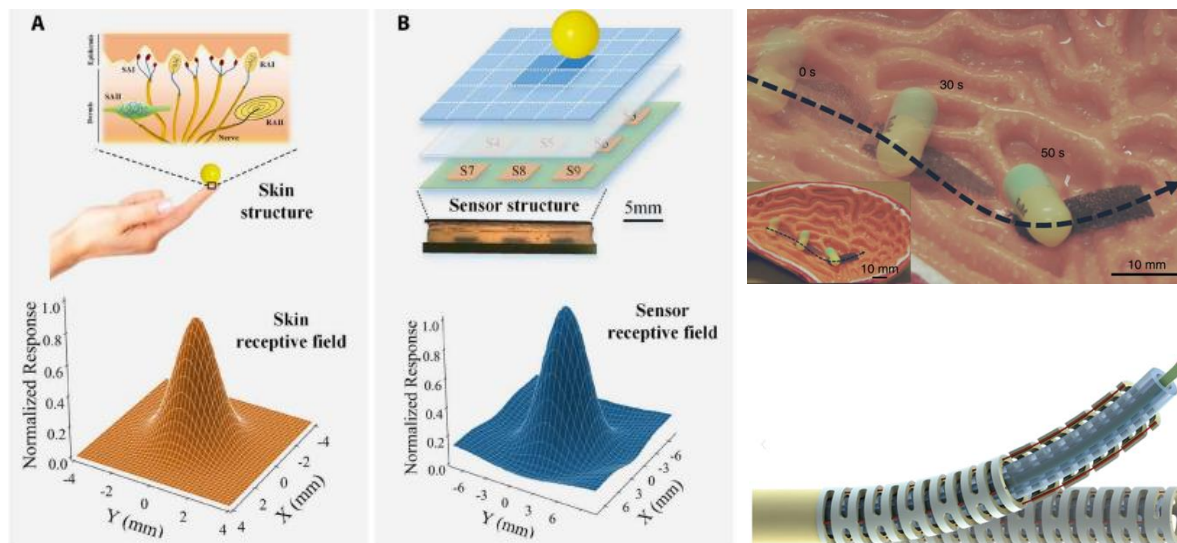
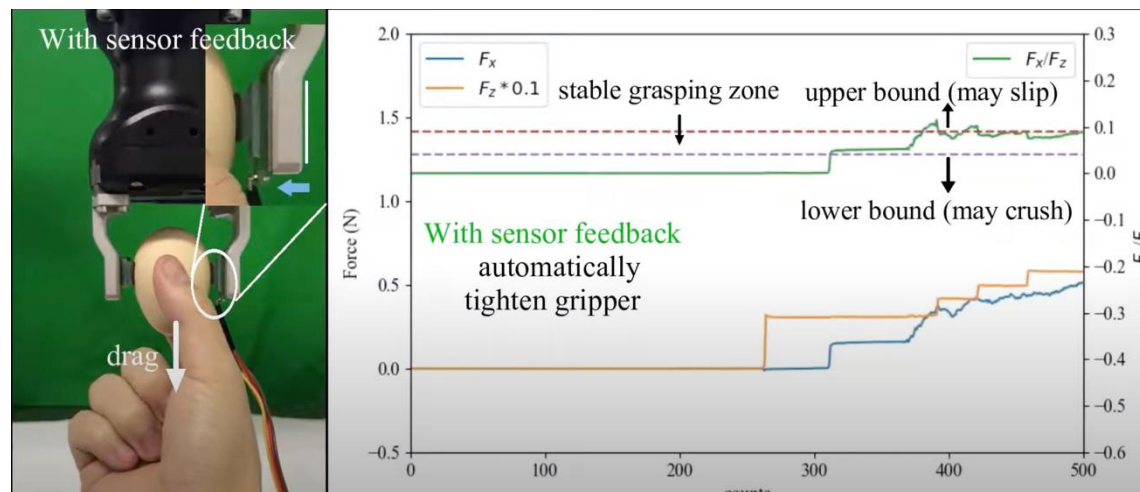
Research Interest:

- Robotics
- Robot manipulation
- Human-robot interaction

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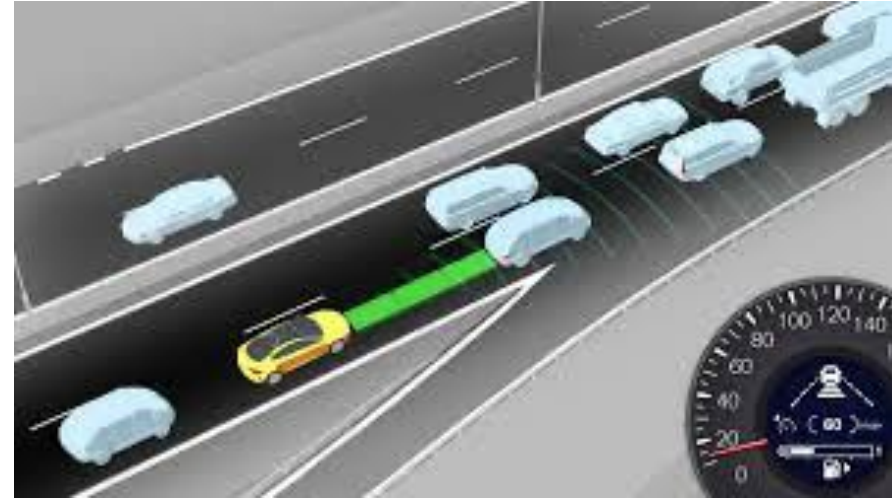


Today's Lecture

- Introduce basic feedback control systems
- Course administration

What is a Control System?

- Tesla Autopilot, Volvo Adaptive Cruise Control, etc



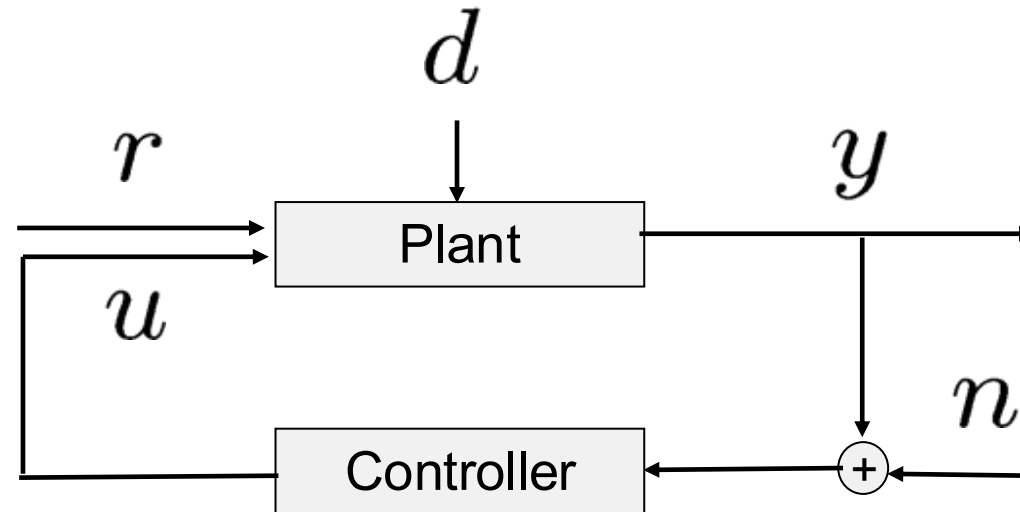
- Andrea Raff's TED talk

The astounding athletic power of quadcopters

<https://www.youtube.com/watch?v=w2itwFJCgFQ>



What is a Control System?



r : reference input

u : controller input

y : sensor output

d : plant disturbance

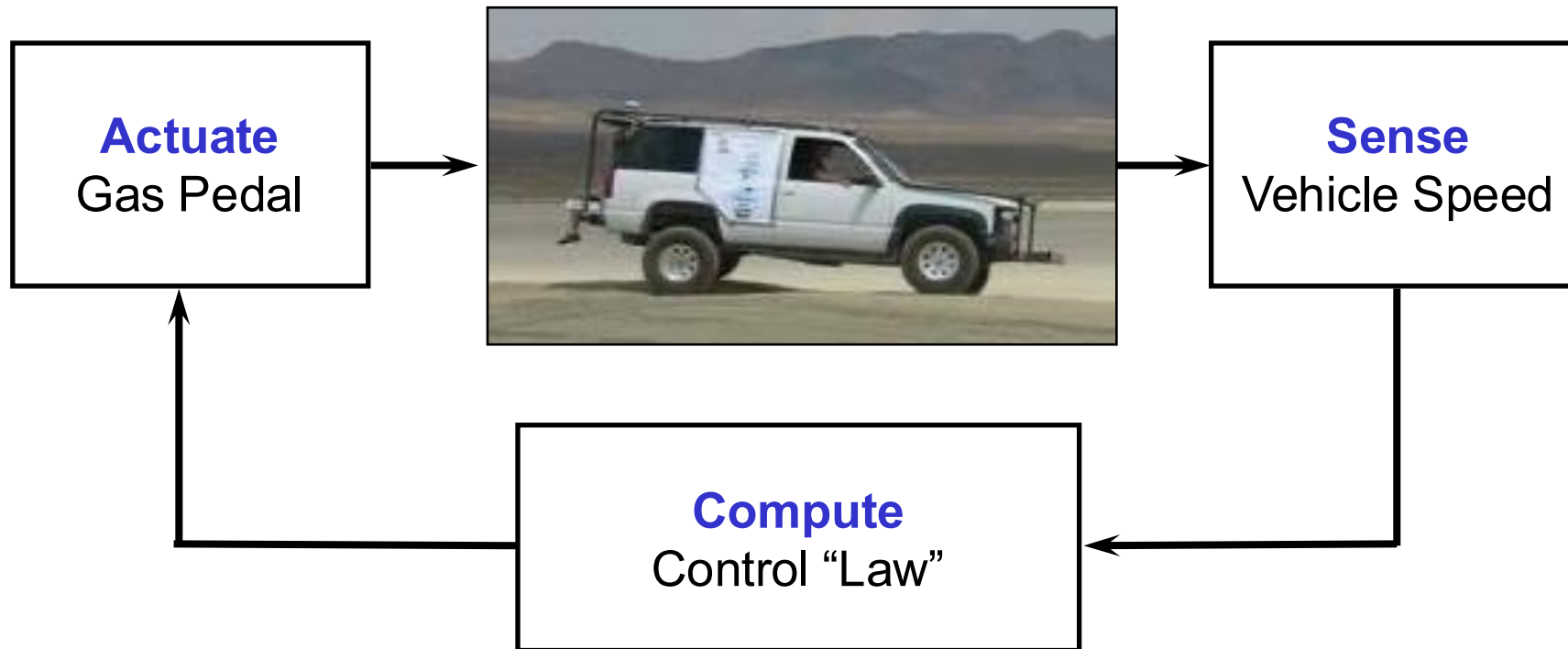
n : sensor noise (or communication noise)

Q: identify the various components in the previous two examples.

Q: a student walks from UG Hall 3 to LTA. identify the various components as a control system.

What is a Control System?

DAPAR Grand Challenge

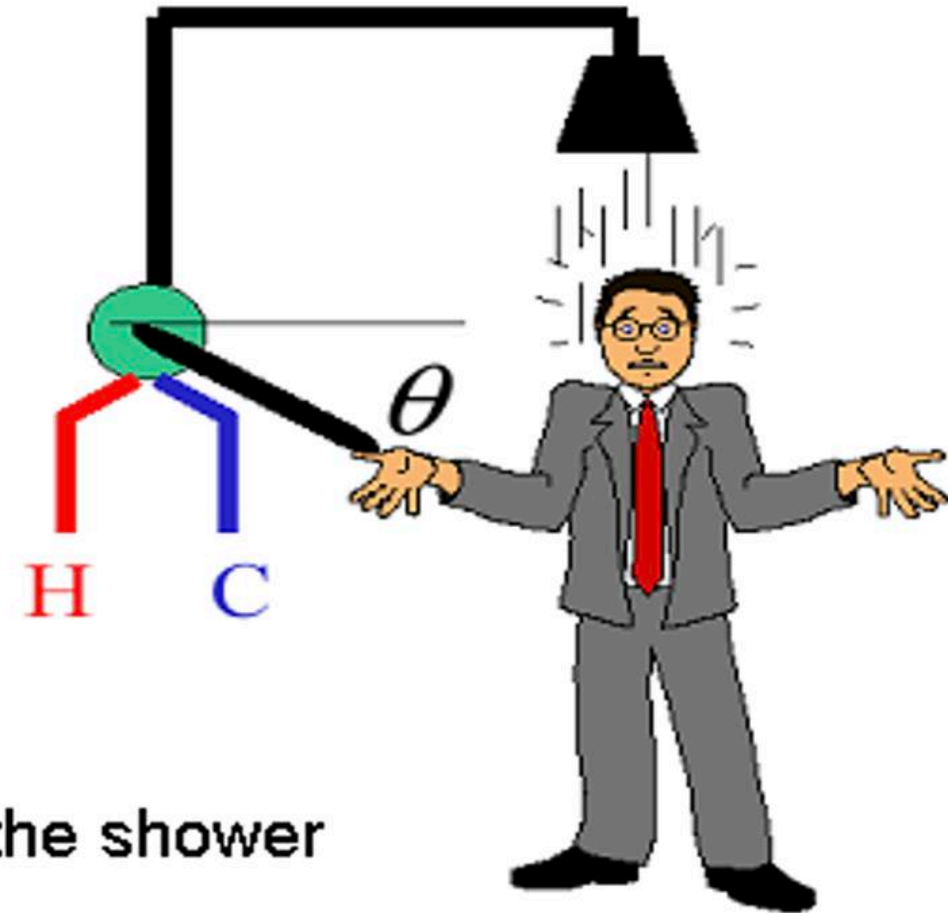
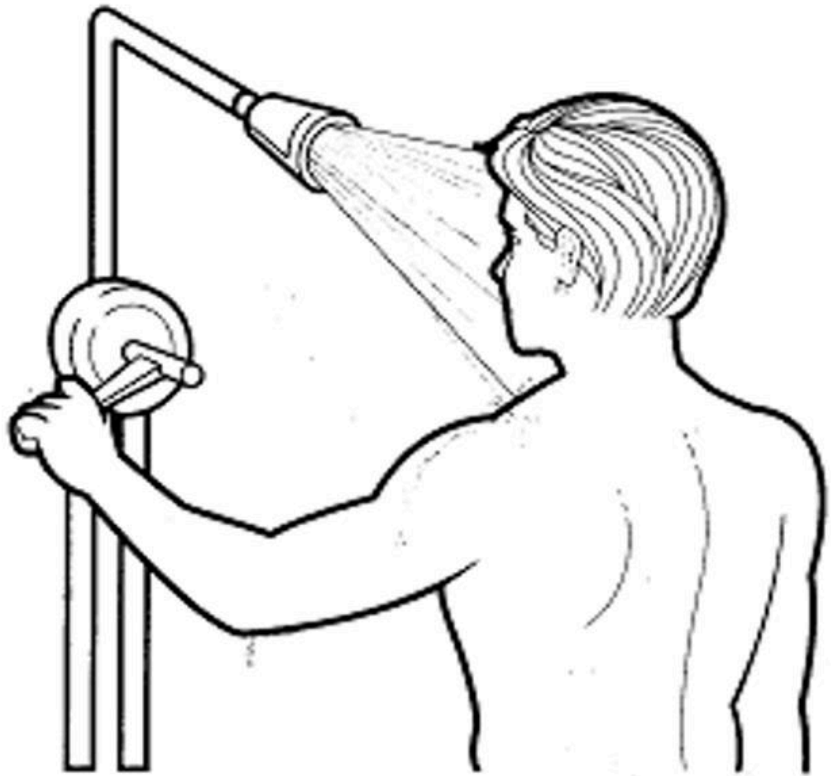


Control = Sensing + Computation + Actuation

In Feedback "Loop"

Examples

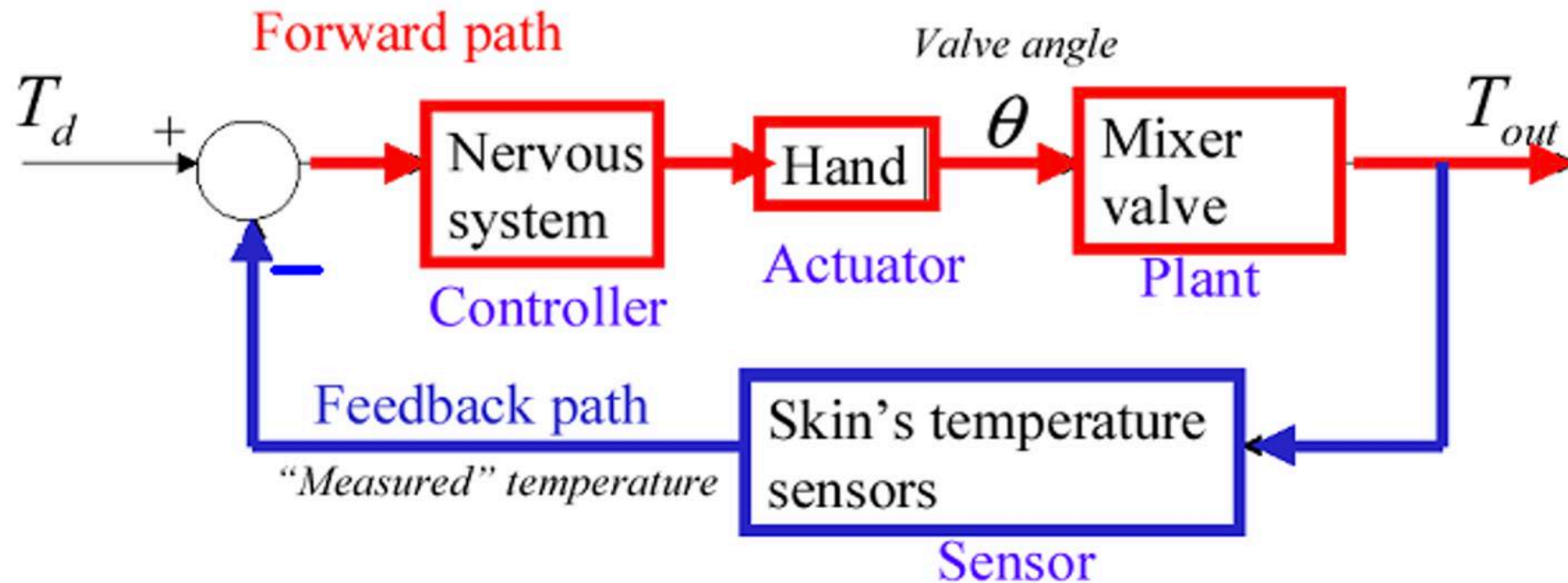
What is a Control System?



In the shower

Examples

Block Diagram Representation



What is a controller?

- A controller is simply a computation unit that computes the “optimal” or “desired” input to the plant.
- One objective of control is to make the plant **stable** and **track** a given reference signal as precisely as possible (e.g., an airplane follows a given trajectory in a stable way).



Open and Closed-loop Control

- Two types of control methods
 - **Open-loop control** – move the robot with a pre-determined way



Q: Can we stabilize the robot with open-loop control?

walk with your eyes closed

Open and Closed-loop Control

- Two types of control methods
 - **Close-loop control**: use the output (i.e. the location of the robot) to adjust the input (i.e. the direction and may be speed of the movement of the robot)
 - We also call it feedback control, since we make the control decision based on the output feedback.



walk with your eyes open

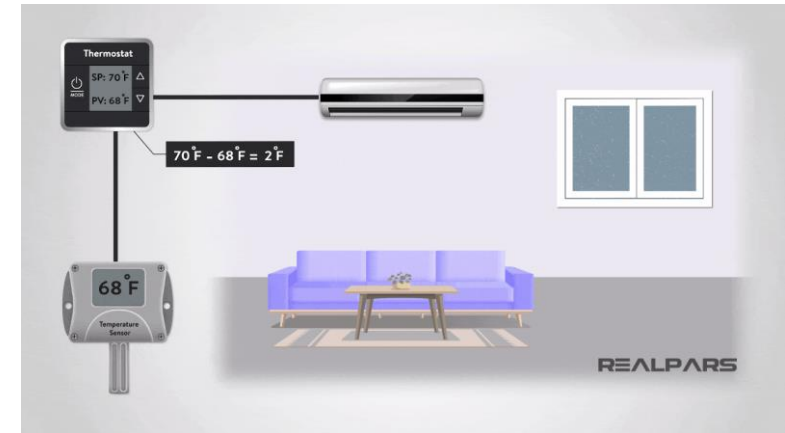
Q: Can we stabilize the robot with closed-loop control?

A: It's possible, but depends on many factors

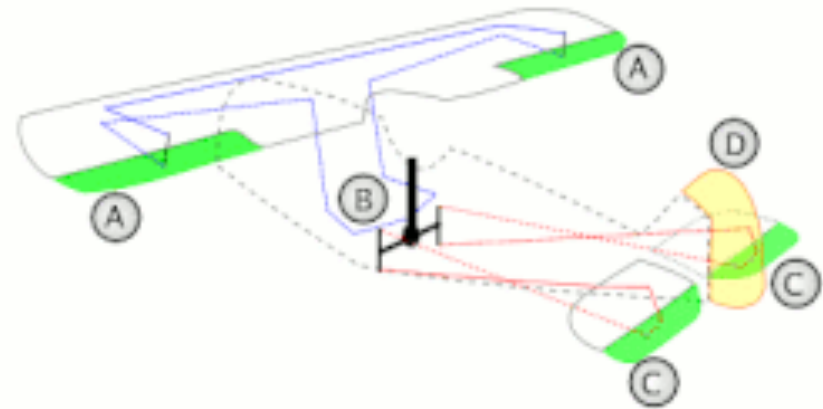
Q: Can you list a few such factors?

What's a Good Control System?

- Make the system **stable** (always the first thing to consider !)
- Be able to **track** a given command
- Be able to **reject disturbance**: e.g., there's a strong wind for the inverted pendulum or for the airplane during take-off
- Be able to **tolerate uncertainty**: e.g., the exact mass is not known



Temperature control



Airplane control

Objectives of This Course

On successful completion of this course, you are expected to be able to:

1. **Model** simple dynamical systems using differential equations and define the role of such models in system analysis, synthesis and simulation.
2. Manipulate **models of LTI systems** in different forms, such as differential equations, transfer functions, and block diagrams, analytically and through the use of CAD tools such as MATLAB.
3. Explain the role of **feedback** as a ubiquitous tool to control a system.
4. Assess the impact of **stability** in a physical system and the use of feedback control to achieve stability.
5. Complete a real **feedback control task** from modeling, controller design, simulation, and implementation using CAD tools such as MATLAB.
6. Identify **robustness and performance** issues in a control system and methods of addressing these issues.

A Tentative Roadmap

Modelling

- Concept of Control and Feedback
- State Space
- System Modelling
- Linearization
- Laplace Transform
- Transfer Function
- Block Diagram
- Simulation and Implementation
- Closed-loop System

Lecture 02-11

Analysis

- Concept of Stability
- Routh Criterion
- Pole Placement
- Bode's Sensitivity
- PID and System Type
- Second-order System
- Extra and Dominant Poles and Zeros
- Case Study

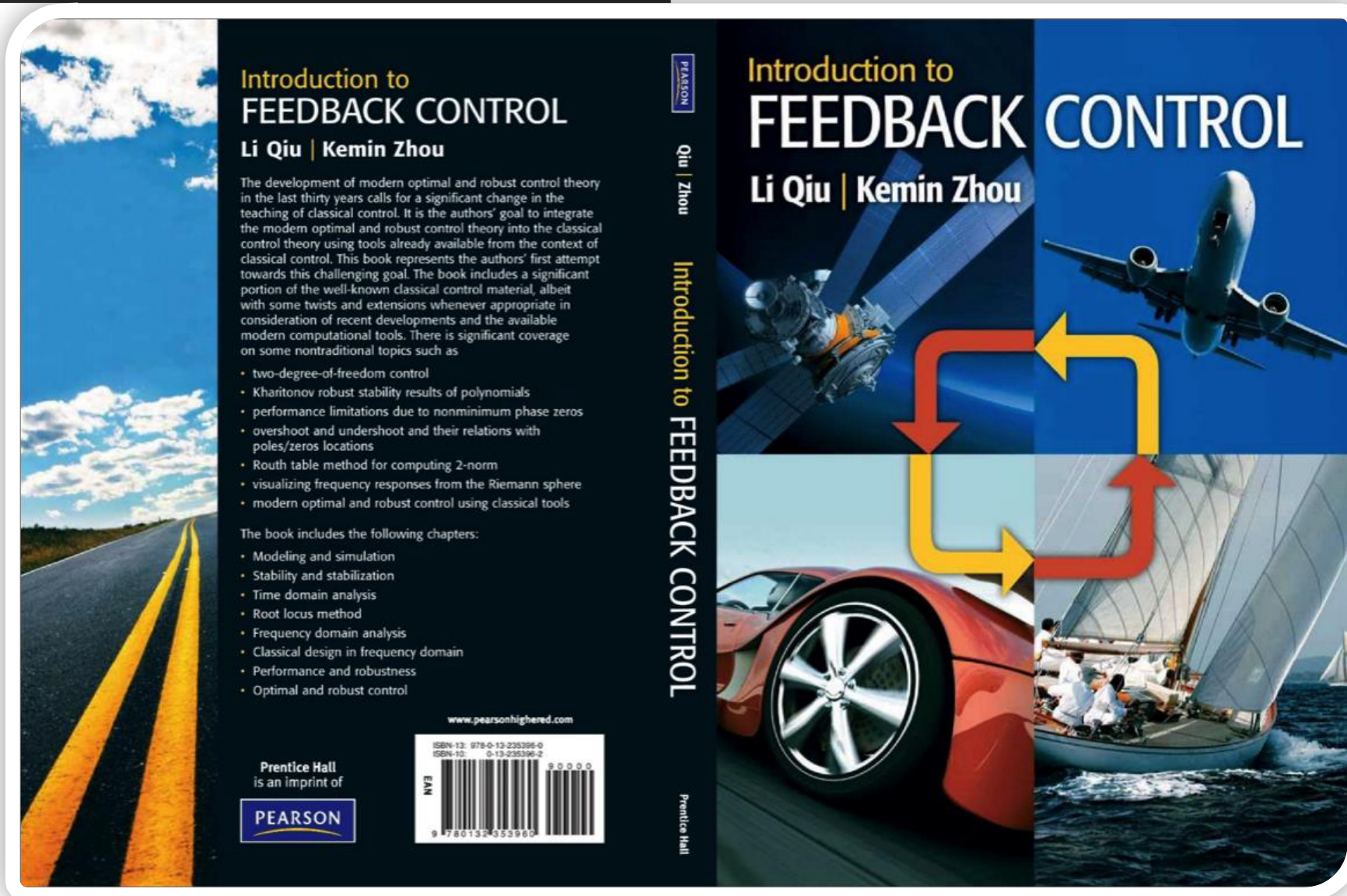
Lecture 12-19

Control

- Root Locus methods
- Advanced control methods*

Lecture 20-23

Textbooks and Lectures



Textbooks and Lectures

- Major Text: uploaded to the Canvas. “Introduction to Feedback Control”, by Li Qiu and Kemin Zhou.
- The set of PPTs I’ll use in the class are mainly contributed by Prof. Li Qiu (and his PhD student Chao Chen at HKUST) and Prof. Maxim Raginsky (UIUC)
- Major Reference:
 - K. Astrom and R. M. Murray, “Feedback systems: an introduction for scientists and engineers”, (softcopy can be found at http://www.cds.caltech.edu/~murray/amwiki/Main_Page)

Grading Policy

- Homework: 20% [4 sets, each 5%]
- Mid-term exam: 30%
- Final exam: 50%
- Labs: 5% [2 sets, each 2.5%] (Bonus, not compulsory)
- To ensure fairness for students who submit assignments on time, a **penalty for late submission** is listed as follows:
 - Late submission within 12 hours, 25% penalty will be applied.
 - Late submission between 12 to 24 hours, 50% penalty will be applied.
 - Late submission for more than 24 hours will not be accepted.
- Any form of cheating (e.g., copy others' homework solutions, lab reports, misconduct in the exam) has absolute ZERO tolerance and will lead to failing the course immediately

Class Schedule

Weeks	Wed (13:30-14:50)	Fri (13:30-14:50)	Note
1	(Feb 4) L01	(Feb 6) L02	
2	(Feb 11) L03	(Feb 13) L04	
3	(Feb 18) No course	(Feb 20) L05	
4	(Feb 25) L06	(Feb 27) L07	Homework1 release after L06
5	(Mar 4) L08	(Mar 6) L09	
6	(Mar 11) L10	(Mar 13) L11	Homework2 release after L11
7	(Mar 18) L12	(Mar 20) L13	
8	(Mar 25) L14	(Mar 27) L15	
9	(Apr 1) Midterm test	(Apr 3) Midterm break	Midterm (in class, open book) [L1-L15]
	(Apr 8) Midterm break	(Apr 10) L16	
10	(April 15) L17	(Apr 17) L18	Lab, Homework3 release after L17
11	(April 22) L19	(Apr 24) L20	Lab
12	(April 29) L21	(May 1) No course	
13	(May 6) L22	(May 8) L23	Homework4 release after L22

- Homework deadline: 11:59pm two weeks after releasing.
- Lab will be tentatively arranged in week 10 and 11.

Class Schedule

L01: Course Introduction

L02: Concept of Control and Feedback

L03: State Space

L04: System Modeling

L05, 06: Linearization

L07: Laplace Transform

L08: Dynamic Response

L09: Transfer Function and Block Diagrams

L10: Simulation and Implementation of Systems

L11: Closed-loop Systems

L12, 13, 14: Stability (Routh Criterion and Pole Placement)

L15: Case Studies

L16: Feedback Control and Bode's Sensitivity

L17: PID and System Type

L18: Second-Order Systems

L19: Extra and Dominant Poles and Zeros

L20, 21, 22: Root Locus Method

L23: Advanced Topics & Review

Modeling

Analysis

Control

Class Expectations

Caution: This course is heavily math-oriented and theoretical. It's also considered to be quite difficult and not easy to get good grade. Thus double think if you really want to enroll this course or not. If you do want to take it, please:

- Attend lectures/labs on time
- Be active during lectures: ask questions
- Work on the homework by yourselves
- Seek help from your classmates or TAs or me if you encounter any difficulties (never keep any problems/questions till the end)

Teaching Team

- **Instructor**

- Prof. Yajing Shen
- Office: Room 2450
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Office hour: by appointment

Advertisement

We currently have projects in the following areas:

- Robot manipulation
- Soft tactile sensor
- AI-enhanced robot control
- Dexterous robot hand
- Biomedical robot

If you are interested and have necessary background, please send me your **transcripts** (UG and PG) & **CV** to eeeyajing@ust.hk together with **your available times**. Please also highlight your area of expertise (especially if you have robotics competition experience). Groups are welcome. This can be **UROP projects** or other forms.

Questions ?



In-class Exercise

1. Draw the curve of $y=x$, $y=x^2$, $y=\cos(x)$, $y=\sin(x)$, $y=e^x$, $y=e^{-x}$

2. Draw the curve of $y=xe^{-x}$ when $x \geq 0$

3. Calculate $\lim_{x \rightarrow \infty} \cos(x) e^{-x}$, $\lim_{x \rightarrow \infty} x^2 e^{-x}$ Ans: 0

4. Calculate y : $\frac{dy}{dx} = y, y(0) = 1$ Ans: e^x

5. Given Taylor series: $f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$

5.1 Proof: $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$

5.2 verify: $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

Take $x_0 = 0$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

6. Given partial integration: $\int u dv = uv - \int v du$

6.1 Proof: $\int x \cos(x) dx = x \sin(x) + \cos(x) + C$

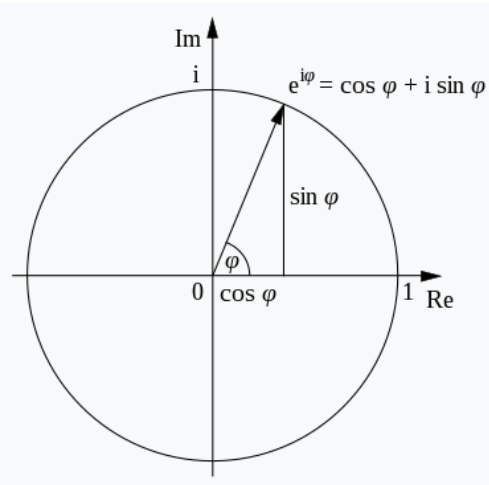
6.2 Proof: $\int x e^x dx = x e^x - e^x + C$

7. Given Euler's formula: $e^{ix} = \cos x + i \sin x$

7.1 Proof:
$$\cos x = \operatorname{Re}(e^{ix}) = \frac{e^{ix} + e^{-ix}}{2},$$
$$\sin x = \operatorname{Im}(e^{ix}) = \frac{e^{ix} - e^{-ix}}{2i}.$$

7.2 Proof: $e^{i\pi} + 1 = 0$

7.3 Proof: $\cos n\theta + i \sin n\theta = (\cos \theta + i \sin \theta)^n$



8. Change it to the form of Matrix, and then calculate the value $\mathbf{x}$ if there is a solution.

$$\text{Q 1: } \begin{cases} x_1 - x_2 + x_3 = 1, \\ x_1 - x_2 - x_3 = 3, \\ 2x_1 - 2x_2 - x_3 = 5. \end{cases}$$

$$\text{Ans: } \begin{cases} x_1 = x_2 + 2 \\ x_3 = -1. \end{cases} \quad \text{Infinite solutions}$$

$$\text{Q 2: } \begin{cases} x_1 - x_2 + x_3 = 1, \\ x_1 - x_2 - x_3 = 3, \\ 2x_1 - 2x_2 - x_3 = 3. \end{cases}$$

$$\text{Ans: } \begin{cases} x_1 - x_2 + x_3 = 1, \\ x_3 = -1, \\ 0 = -2. \end{cases} \quad \text{No solution}$$

9. Find the value of a to make the equations have solution $\mathbf{x}$:

$$\begin{cases} 3x_1 + x_2 - x_3 - 2x_4 = 2, \\ x_1 - 5x_2 + 2x_3 + x_4 = -1, \\ 2x_1 + 6x_2 - 3x_3 - 3x_4 = a + 1, \\ -x_1 - 11x_2 + 5x_3 + 4x_4 = -4 \end{cases} \quad \text{Ans: } a=2$$

10. If a solution exists, determine whether there is a unique or infinite solution.

Ans: Suppose the number of non-zero rows is r , the number of unknowns is n . If $r=n$, there is a unique solution. If $r < n$, there are infinitely many solutions.